

The Curse of the Mandate of Heaven: A Model of Power Centralization

Shuyi Yu

University of Chicago

yushuyi@uchicago.edu

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Abstract

Europe and China followed divergent paths of development during the premodern period. Europe nurtured representative institutions and secure property rights that facilitated the rise of modern capitalism, whereas the growth of commercial interests in China was persistently stifled. The divergence can be attributed, at least in part, to the more centralized power structure of the Chinese state at the dawn of the premodern era. To account for this distinction, I propose that China's early adoption of the Mandate of Heaven, a merit-based political philosophy aimed at legitimizing imperial rule in a secular society, inadvertently contributed to the cyclical overturning of dynasties and a steady trend of power centralization towards the emperor. In substantiating this argument, I develop a game-theoretical model of authoritarian power sharing, demonstrating that the threat of peasant revolution can alter the initial power distribution between the ruler and the elite, tilting the equilibrium more in favor of the ruler and leading to a less balanced power structure. I also present historical narratives to corroborate and contextualize this finding.

1. INTRODUCTION

A striking contrast exists in the political development between Imperial China and Medieval Europe. From 200 BC to the 15th century, China underwent multiple cycles of power centralization, resulting in an imperial state where the emperor held nearly absolute authority (Wittfogel, 1957; Hui, 2005; Von-Glahn, 2016; Wang, 2022). In contrast, during the Middle Ages after the decline of the Roman Empire, European states featured a relatively fragmented political landscape, characterized by a dispersion of power among the monarch and the feudal lords (Wickham, 1984; Dincecco, 2010; Hoffman, 2015; Scheidel, 2019; Stasavage, 2020). The influence of these power structures had far-reaching consequences, particularly in shaping the divergent paths of political and economic institutions between China and Europe as they entered the modern era (Acemoglu et al., 2005a,b; Acemoglu and Robinson, 2012, 2019; Rosenthal and Wong, 2011; Mokyr, 2016). Among the significant ramifications, the most prominent is perhaps the emergence of representative institutions and secure property rights in Europe, which are widely recognized as fundamental catalysts for the Industrial Revolution (North and Thomas, 1973; North and Weingast, 1989). Hence, the divergence between China and the Western world in the modern era can be attributed, at least in part, to the disparity in power structures. However, the existing literature offers limited insights into uncovering the underlying cause behind this disparity, primarily relying on geographical or environmental arguments. (Wittfogel, 1957; Jones, 1981; Wickham, 1984; Turchin, 2009; Ko et al., 2018; Fernandez-Villaverde et al., 2023).

In this paper, I aim to present a novel explanation for the divergence in political structures between Imperial China and Medieval Europe by proposing a model of authoritarian power sharing. Departing from the existing literature that primarily examines the power struggle between two political entities, such as a ruler and an elite, I analyze how the entry of a third party, i.e. the peasantry, leads to a shift in the original power structure. More precisely, the distinctive characteristic of imperial Chinese politics that peasants possess the legitimacy to overthrow the entire upper class through violence and assume the positions of the ruler and the elite themselves, inadvertently enables the ruler to seize power from the elite and solidify his own supremacy. This result stems from the elite's fear that his defiance against the ruler's meager offer could potentially trigger a complete reshuffling of power, ultimately resulting in his own downfall from current socioeconomic status. Such fear renders the elite inclined to acquiesce to the encroachment of the ruler upon his power, resulting in a less balanced

power structure within the upper class.

Political economists have long recognized the relatively limited absolute power held by European rulers, especially when studying the pivotal role played by representative institutions and secure property rights in the development of the industrial revolution and modern capitalism (North and Weingast, 1989; Acemoglu et al., 2005a,b). In their characterization of inclusive economic institutions, Acemoglu and Robinson (2012, 2019) has identified the weaker autocratic power as a characteristic of “broad distribution of political power.” Historically, this contrast is manifested in the varying degrees of control exerted by Chinese and European rulers over the lands and populations of their respective states, as well as the laws governing the succession of elite titles. In Medieval Europe, control over lands and peasants primarily rested with the overlords, while the kings held a position of “first among equals,” lacking direct control over other territories (Bloch, 1962). As put by Levi (1988), “a monarch was expected to ‘live of his own’ (*vivre du sien*). That is, funds for the monarch were to come from royal lands and customary dues...Should monarchs need more, even if it was to fund a campaign on behalf of the country as a whole, they had to obtain assent to some form of ‘extraordinary’ taxation. They could neither expropriate property at will nor rely on a regular levy.” Meanwhile, Chinese emperors held ultimate control over the entire territory and population of the country. Land re-centralization and land confiscation were common in Imperial China (Chao and Chen, 1982; Ebrey and Walthall, 2013; Zhang, 2015; Von-Glahn, 2016; Wang, 2000), where the entire populace was regarded as “subjects” of the emperor (Finer, 1997a), granting him the authority to dispense rewards and punishments capriciously (Finer, 1997b). Moreover, in Medieval Europe, noble titles were hereditary, and the monarch wielded only indirect influence over their succession (Finer, 1997b); whereas in Imperial China, scholar-bureaucrats were selected through imperial examinations overseen by the emperor and his appointees, resulting in non-hereditary titles subject to arbitrary abolishment (Levenson, 1965).

However, while there has been extensive research examining the socioeconomic implications resulting from the divergence of power structures between Imperial China and Medieval Europe, there remains a relative scarcity of literature dedicated to exploring the underlying origins of this phenomenon. Some studies argue that a centralized *de facto* power can further reinforce itself by stipulating *de jure* rules advantageous to its own interests, and *vice versa*. Historical accounts suggest that due to their initial

weak position, European monarchs had to cede more political power to feudal lords in exchange for financial support during times of war, which further undermined their authority (Wickham, 1984; Bates and Lien, 1985; Levi, 1988; North and Weingast, 1989; Stasavage, 2011). Whereas the initial political centralization in China effectively restricts the available exit options for elites, placing them at a disadvantage during negotiations with the ruler, thereby facilitating the consolidation of power in the ruler's hands (Hui, 2005; Dincecco and Wang, 2018). More relevant to my paper, Jia et al. (2021) constructs a model illustrating how the powerful emperors in Imperial China can further enhance their authority by raising the benefits provided to the general populace to match those of the elites, thereby securing support from the former when facing challenges from the latter. However, this line of reasoning assumes an initial inequality in power distribution, which requires further justification. Another significant strand of literature examines the divergent geographical conditions faced by Imperial China and Medieval Europe (Wittfogel, 1957; Jones, 1981; Wickham, 1984; Turchin, 2009; Ko et al., 2018; Fernandez-Villaverde et al., 2023). Nevertheless, ascribing the divergence solely to random factors may appear insufficient and unsatisfying in providing a comprehensive understanding of human history and offering valuable lessons for human activities.

So what serves as the starting point for contemplating the Great Divergence between Imperial China and Premodern Europe? While it is a reasonable assumption that most rulers would seek to consolidate power in their own hands, why were Chinese emperors successful in achieving this while their European counterparts were not? All of these inquiries likely converge on the fundamental question of political legitimacy. Under the feudal system, European kings and lords primarily derived their legitimacy from the Christian church and through mutual recognition, often facilitated by marriage alliances. In contrast, China has upheld a predominantly secular society since its inception, lacking a higher authority to bestow legitimacy upon Chinese emperors. Nevertheless, the Chinese devised an alternative approach to address this matter. As early as the Zhou dynasty (1046-256 BC), Chinese emperors crafted the political philosophy known as the "Mandate of Heaven" to confer upon themselves the ultimate authority to govern. This philosophy embodies a meritocratic essence, endowing rulers who deliver effective governance with the mandate to rule, while simultaneously affording all subjects, regardless of their social status, the right to rebel in the event of unjust or inept rule. In accordance with this philosophy, peasants actively participated in the political arena throughout Chinese history,

starting from an early age and on a large scale. They initiated numerous rebellions, successfully overthrowing several regimes, and even established a few dynasties¹. In that sense, the Chinese political system was remarkably progressive compared to Europe, where commoners began to emerge in political life only around the 1600s after acquiring sufficient economic power through business and trade (Acemoglu et al., 2005a,b), a staggering 1800 years later than in China. Certainly, my discussion is confined to the historical periods of Imperial China and Medieval Europe, as the legitimacy of mass rebellion has gained widespread acceptance in the Western world during the modern era² (Acemoglu and Robinson, 2000, 2001).

Here, I propose a repeated game to demonstrate how the political legitimacy of the peasant revolution can significantly impact the power structure. In this model, there are three actors: a ruler, an elite, and a peasant hero, the latter potentially drawn from a population of peasants. All three actors are strategic, and the ruler and the elite constitute the upper class. In each period, the ruler and the elite bargain over the division of power, which is modeled as a flow of rents normalized to 1. The ruler moves first and proposes the division, but is subject to the elite's threat to challenge and replace him. In the absence of the peasants, the ruler and the elite tend to reach a peaceful equilibrium where power distribution is determined by the credibility of the elite's threat. However, when a peasant hero is allowed to arise and given the choice to overthrow the entire upper class if the latter is divided, a new peaceful equilibrium emerges in which the ruler shares less power with the elite. This shift occurs because the threat of a peasant revolution and a potential complete reshuffling of power put pressure on the elite to accept a less advantageous peace agreement proposed by the ruler. Consequently, the ruler can exploit the first mover advantage and further centralize power. What's more, the peasant hero finds it more profitable to become a ruler and would indeed choose to launch a revolution when given the choice, thus validating the credibility of its entry threat. This dynamic establishes a self-fulfilling equilibrium.

My theory provides a novel perspective on authoritarian politics by investigating how the introduction of a third party disrupts traditional power-sharing dynamics. In the existing literature, authoritarian

¹Notable examples include the Han dynasty (206 BC-220) and the Ming dynasty (1368-1644).

²Exemplified by the French Revolution (1789-1799).

politics are often portrayed as power struggles between two primary actors: the ruler versus the winning coalition in power-sharing models (Svolik, 2009; Meng et al., 2023), and the elite versus the poor in regime change models (Acemoglu and Robinson, 2000, 2001). These models usually culminate in an absorbing state where either the ruler consolidates power or the poor initiate a revolution, thereby transitioning the regime into a democracy. My model extends these frameworks to encompass all three key roles: the ruler, the elite, and the peasantry, as strategic players, and it examines the strategic interplay between the threats posed by both the elites and the peasantry to the ruler. Recent research highlights the strategic complementarity between elites and citizens in collective actions (Mesquita, 2010; Casper and Tyson, 2014; Garfias and Sellars, 2022), where elites typically do not suffer and may even gain from the regime’s collapse, thus fostering a natural alliance between elites and citizens in opposition to the ruler. The challenge that remains is to ensure effective coordination between and within these two opposition groups under conditions of incomplete information. However, in Imperial China, dynastic changes brought by peasant revolutions usually resulted in brutal purges and the violent replacement of the existing elites by ascending peasants. This instilled more fear among the elites regarding potential regime changes than the prospect of losing power to the ruler. Consequently, popular uprisings inadvertently enhance the ruler’s grip on power, a departure from the existing literature. Another deviation from traditional models is the absence of an absorbing state in my approach. When either the elite or the peasant hero emerges as the ultimate victor, they assume the role of the ruler and continue the game with a new elite. This approach seems more suitable for modeling the cyclic patterns of historical circumstances in Imperial China.

The study offers valuable insights into the emergence of specific institutions in Chinese history. One notable example is the imperial examination system, which originated during the Sui Dynasty (581-618), underwent further development in the Tang Dynasty (618-907), was solidified during the Song Dynasty (960-1279), and persisted until the late Qing Dynasty (1636-1905). It was one of the earliest institutional innovations in human history to select bureaucrats based on merit rather than birth. According to my model, the introduction of the examination system can be explained as a strategic move by the ruler to seize power from the elites. As expected, this reform encountered strong opposition from the aristocrats throughout the Sui and Tang dynasties. The eventual consolidation of the examination system in the Song dynasty can be attributed, in part, to the tumultuous period known as

the Five Dynasties and Ten Kingdoms (907-960). This chaotic era was marked by continuous warfare among warlords of commoner backgrounds who aspired to become emperors, leaving the aristocracy in a vulnerable position. Subsequently, the surviving old aristocrats and newly formed aristocrats were more amenable to accepting the emperor's reforms. This observation lends support to my theory that the looming threat of commoner violence compelled the elites to relinquish power to the ruler. My theory aligns with [Jia et al. \(2021\)](#) in that the examination system aims to reduce the asymmetry between the elites and the common people, ultimately consolidating power in the hands of the ruler. However, in my model, this is achieved by diminishing the power of the elites rather than elevating the power of the commoners. Moreover, my theory indicates that institutional changes are the consequences of power dynamics, setting it apart from other research perspectives that consider the examination system and bureaucracy as external factors influencing power consolidation in imperial China ([Stasavage, 2020](#); [Wang, 2022](#)).

The study contributes to the broader exploration of the influence of war on political development. Scholars have extensively investigated the effects of prolonged inter-state wars in Europe on state building and power distribution within states. Research in this area has revealed a correlation between inter-state warfare and the evolution of political representation and parliamentary politics in European states ([Bates and Lien, 1985](#); [Levi, 1988](#); [Tilly, 1992](#); [Ertman, 1997](#); [Stasavage, 2011](#); [Blaydes and Chaney, 2013](#); [Downing, 2020](#)). Despite similar levels of engagement in warfare, China has followed a different trajectory ([Hoffman, 2015](#)). Historical evidence indicates that conflicts with northern nomads, as well as internal wars and rebellions, played a role in facilitating power centralization throughout Chinese history ([Turchin, 2009](#); [Wang, 2016, 2017](#); [Ko et al., 2018](#); [Dincecco and Wang, 2018](#); [Wang, 2022](#)). Scholars have attempted to reconcile the contradictory observations by differentiating between the types of wars fought, the recruitment strategies employed, the technologies utilized, and the methods of financing involved ([Scheve and Stasavage, 2010, 2012, 2016](#); [Hui, 2005](#); [Blaydes and Chaney, 2013](#); [Hoffman, 2015](#); [Dincecco and Wang, 2018](#)). My paper seeks to offer a novel interpretation of the differing impacts of wars on power structures, suggesting that the distinguishing feature between peasant rebellions in Imperial China and their counterparts in Medieval Europe may be their legitimate claims rather than merely their presence. My paper is particularly relevant to studies on the political centralizing effects of peasant rebellions in Imperial China. [Wang \(2017\)](#) provides evidence

that the Qing emperors capitalized on the Taiping Rebellion (1850–1864) to enhance their coercive power and the gentry class provided substantial financial support to these military endeavors. Wang (2022) examines the Huang Chao Rebellion (874–884) during the Tang Dynasty and proposes that the rebels’ physical elimination of the old elites led to a power imbalance in the subsequent Song Dynasty (960–1279), which endured in the long term. Nonetheless, the course of Chinese history demonstrates a consistent and gradual strengthening of power centralization, which necessitates a systematic explanation that extends beyond the experience of particular dynasties. In this regard, my model employs a repeated framework that allows for the consideration of the persistent threat posed by peasants and the evaluation of its influence on the evolving power structure.

The paper is structured as follows: In Section 2, I present the model and discuss the equilibrium analysis. In Section 3, I graph the parametric space and discuss the potential path of equilibrium transitions. In Section 4, I delve into historical evidence in Imperial China, drawing connections to the specific mechanisms of the model. Section 5 concludes the paper with implications for different historical settings.

2. A MODEL OF POWER CENTRALIZATION

2.1. Setup

Consider a polity that produces a flow of rents 1 at each $t = 0, 1, 2, \dots$. At each t , the polity has a *ruler*, an *elite*, and a population of peasants. The *upper class* of the polity consists of the ruler and the elite. Each player is forward looking and shares a common discount factor $\delta \in (0, 1)$.

At the beginning of period t , the ruler allocates the rents within the upper class, sharing $s_t \in [0, 1]$ to the elite and preserving the rest $1 - s_t$. Then, the elite may get an opportunity to *challenge* the ruler, with the purpose of replacing the ruler. Let $\theta_t \in \{0, 1\}$ represent whether the elite has such an opportunity and

$$\Pr(\theta_t = 1) = \pi,$$

where $\pi \in (0, 1)$. After getting such an opportunity, $\theta_t = 1$, the elite chooses whether to challenge the ruler, $c_t \in \{0, 1\}$. At last, a *peasant hero* may be randomly drawn from the population of the peasants,

who is able to *revolt*, with the purpose of purging the upper class. Let $h_t \in \{0, 1\}$ represent whether a peasant hero arises and

$$\Pr(h_t = 1) = \theta_t c_t \phi,$$

where $\phi \in (0, 1)$. In other words, peasant revolution is possible only when the upper class is *divided*, that is, when the elite attempts to replace the ruler, then with probability ϕ , a peasant hero arises. After a peasant hero is drawn, $h_t = 1$, the peasant hero chooses whether to launch a revolution, $r_t \in \{0, 1\}$.

Period t may end with three possibilities: *peace*, *internal conflict*, and *revolution*. First, it would end with peace if $\theta_t c_t = 0$, that is, either if the elite has no opportunity to challenge the ruler or if the elite has such an opportunity yet chooses not to challenge the ruler. In this case, both the ruler and the elite survive to the next period and keep their positions. Second, period t would end with an internal conflict if $\theta_t c_t = 1$ and $h_t r_t = 0$, that is, if the elite challenges and no revolution is launched, either because there is no peasant hero or because the peasant hero chooses not to revolt. In this case, either the ruler or the elite prevails and permanently eliminates the other. The loser gets 0 at the current and all subsequent periods, while the winner gets all the rents at the current period and becomes the ruler at the next period, sharing the rents with a new elite. The probability for the elite to defeat the ruler in an internal conflict is $p \in (0, 1)$. Third, period t would end with a revolution if $\theta_t c_t = 1$ and $h_t r_t = 1$, that is, if the elite challenges the ruler and a peasant hero launches a revolution. In this case, both the ruler and the elite are eliminated and both get 0 at the current and all subsequent periods. The peasant hero takes all the rents at the current period and becomes the ruler at the next period, sharing the rents with a new elite. In other words, the peasant hero and the new elite would form the new upper class after revolution. However, launching a revolution imposes on the peasant hero a lump-sum cost of $k > 0$.

The stage game at t has the following sequence of moves. The solution concept is *equilibrium in stationary strategies*. The subscript “ t ” is omitted whenever it raises no confusion.

1. The ruler chooses $s_t \in [0, 1]$.
2. $\theta_t \in \{0, 1\}$ is drawn.

3. The elite observes θ_t and, if $\theta_t = 1$, chooses $c_t \in \{0, 1\}$.

4. $h_t \in \{0, 1\}$ is drawn.

5. If $h_t = 1$, the peasant hero chooses $r_t \in \{0, 1\}$.

Table 1 lists all the symbols used in the paper for quick reference.

Table 1: Symbols

Category	Symbol	Substantive Meaning
<i>background</i>	$t \in \{0, 1, 2, \dots\}$	time
	$\delta \in (0, 1)$	common discount factor
<i>ruler</i>	$s_t \in [0, 1]$	ruler's decision: rents shared with elites
<i>elite</i>	$\theta_t \in \{0, 1\}$	whether elite can challenge ruler
	$\pi \in (0, 1)$	probability of $\theta_t = 1$
	$c_t \in \{0, 1\}$	elite's decision: whether to challenge ruler
	$p \in (0, 1)$	probability to defeat ruler given $c_t = 1$
<i>peasant hero</i>	$h_t \in \{0, 1\}$	whether a peasant hero arises
	$\phi \in (0, 1)$	probability of $h_t = 1$ given $c_t = 1$
	$r_t \in \{0, 1\}$	peasant hero's decision: whether to revolt
	$k > 0$	peasant hero's cost of revolting

2.2. Comment on the Primitives

It is useful to briefly discuss an important assumption of this model, which will play an important role in the rest of my analysis. I assume that the ruler makes the rents-sharing decision prior to the realization of the threat from the elite. I believe that this is often a realistic assumption in real life, as institutional choices regarding rent-sharing are typically made at the beginning of a period and will persist until the emergence of opportunities for elites' challenges. However, this assumption departs from [Acemoglu and Robinson \(2001\)](#), where the opportunity cost of the revolution, i.e. their state variable, is revealed before the rich set the tax rate. While incorporating their timeline wouldn't compromise my fundamental logic, I opt to forgo it for the sake of tractability.

2.3. Equilibrium Analysis

Consider any given equilibrium. Two quantities are crucially important in characterizing the equilibrium: s^* as the equilibrium amount of rents the ruler shares to the elite, which inversely measures the *degree of power centralization* by the ruler, and ϕ^* as the equilibrium probability that a peasant

revolution occurs when the upper class is divided, which measures the strength of the *threat of peasant revolution*. To facilitate subsequent analysis, also let V^* denote the equilibrium discounted payoff of the ruler and U^* be that of the elite.

2.3.1 Peasant Revolution

First, consider the case when $\theta c = 1$ and $h = 1$, that is, when the elite challenges the ruler, so that the upper class is divided, and a peasant hero is drawn. The total discounted payoff the peasant hero gets by launching a peasant revolution is

$$1 + \delta V^*.$$

Recall that through a revolution, the peasant hero eliminates the upper class, getting 1 in the current period, and becomes the new ruler, getting V^* in the future periods. The peasant hero would choose to launch a revolution, $r = 1$, if and only if such a payoff justifies the cost k . As a result, the probability that a peasant revolution occurs when the upper class is divided must satisfy

$$(1) \quad \phi^* \begin{cases} = 0, & 1 + \delta V^* < k \\ \in [0, \phi], & 1 + \delta V^* = k \\ = \phi, & 1 + \delta V^* > k \end{cases}.$$

If $\phi^* > 0$, the upper class — the ruler and the elite — must face a non-trivial threat of peasant revolution in equilibrium. Such a threat imposes on the upper class a risk of being divided. When it is divided, with probability ϕ^* , a peasant hero would rise and eliminate the entire upper class in a peasant revolution. Note in (1) that $\phi^* > 0$ holds if and only if V^* is sufficiently large. This implies that the upper class would have to cope with the threat of peasant revolution when the ruler is sufficiently better off in equilibrium, or becoming a ruler is attractive enough for the peasant hero.

2.3.2 Elite's Challenge

Now consider the case when $\theta = 1$, so that the elite has an opportunity to challenge the ruler, and supposed the ruler has offered s to share with the elite. By choosing to accept the rents the ruler allocates, the elite gets the shared rents s in the current period and remains to be an elite in the next period, expecting to get U in the future periods. The total discounted payoff the elite gets by not

challenging the ruler thus is

$$s + \delta U^*.$$

Suppose the elite chooses to challenge the ruler. First, because the upper class is divided, with probability ϕ^* , a peasant hero would rise to eliminate the entire upper class including the elite. In this case, the elite gets 0. Second, with probability $(1 - \phi^*)(1 - p)$, peasant revolution would not happen and the upper class would be preserved, but the ruler would defeat and eliminate the elite in the internal conflict. In this case, the elite gets 0 as well. The elite can benefit from challenging the ruler only when there is no peasant revolution and, at the same time, the elite defeats the ruler in the internal conflict. This happens with probability $(1 - \phi^*)p$, in which case the elite gets the discounted payoff of

$$1 + \delta V^*.$$

Hence, by challenging the ruler, the elite expects to get in total

$$(1 - \phi^*)p(1 + \delta V^*).$$

Therefore, given the opportunity, the elite would challenge the ruler, $c = 1$, if and only if

$$(1 - \phi^*)p(1 + \delta V^*) > s + \delta U^*.$$

In other words, to buyoff the elite so as to avoid conflict, the ruler has to share

$$(2) \quad s \geq (1 - \phi^*)p(1 + \delta V^*) - \delta U^*.$$

The right hand side of (2) is the smallest amount of rents the ruler must forgo to the elite for peace. Peace is more affordable for the ruler when the elite is better off in equilibrium, so that U^* is larger, and/or when the ruler is worse off, so that V^* is smaller. Both makes challenging the ruler less attractive from the perspective of the elite. Most interestingly, the price the ruler has to pay for peace is lower when the upper class faces a greater threat from peasant revolution, that is, when ϕ^* is larger. The elite has a weaker incentive to challenge the ruler and therefore introduces division from within the upper class when doing so bears a worse risk of a peasant revolution that purges both the elite

and the ruler.

2.3.3 Ruler's Choice

At last, consider the ruler's decision in choosing how much rents to share with the elite. If the ruler shares enough rents with the elite, so that (2) holds, status quo would be maintained regardless of whether the elite has any opportunity to challenge. In this case, the ruler gets $1 - s$ at the current period and keeps being the ruler, getting V^* for the future periods. The total discounted payoff of the ruler therefore is

$$1 - s + \delta V^*.$$

If the ruler shares less, so that (2) fails, there are two possibilities. First, with probability $1 - \pi$, the elite has no opportunity to mount a challenge, $\theta = 0$, so that the ruler gets in total

$$1 - s + \delta V^*.$$

Second, with probability π , the elite gets an opportunity, $\theta = 1$, and therefore challenges the ruler, so that the upper class is divided. In this case, the ruler expects to get

$$(1 - \phi^*)(1 - p)(1 + \delta V^*).$$

The ruler benefits only when there is no peasant revolution that eliminates the entire upper class, which happens with probability $1 - \phi$, and when the ruler defeats the elite in the internal conflict, which happens with probability p . Hence, if the ruler chooses an s that fails (2), the expected total discounted payoff would be

$$\begin{aligned} & (1 - \pi) \overbrace{(1 - s + \delta V^*)}^{\theta=0} + \pi \overbrace{(1 - \phi^*)(1 - p)(1 + \delta V^*)}^{\theta=1} \\ &= (1 - \pi(\phi^* + (1 - \phi^*)p))(1 + \delta V^*) - (1 - \pi)s. \end{aligned}$$

In general, by sharing s , the ruler gets in total

$$(3) \quad \begin{cases} 1 - s + \delta V^*, & s \geq (1 - \phi^*)p(1 + \delta V^*) - \delta U^* \\ (1 - \pi(\phi^* + (1 - \phi^*)p))(1 + \delta V^*) - (1 - \pi)s, & s < (1 - \phi^*)p(1 + \delta V^*) - \delta U^* \end{cases}.$$

The ruler seeks to maximize the above expression in choosing s . Note that (3) is strictly decreasing in s except at $s = (1 - \phi^*)p(1 + \delta V^*) - \delta U^*$. Intuitively, the ruler can buy off the elite to maintain peace by sharing exactly $(1 - \phi^*)p(1 + \delta V^*) - \delta U^*$, so that sharing any rents more than that would be wasteful. Similarly, the ruler has to face the elite's challenge anyway by sharing less than $(1 - \phi^*)p(1 + \delta V^*) - \delta U^*$, so that sharing any amount positive but below that would be inferior to sharing nothing. Hence, in equilibrium, the ruler either shares exactly $(1 - \phi^*)p(1 + \delta V^*) - \delta U^*$ to make the elite indifferent, getting

$$1 - ((1 - \phi^*)p(1 + \delta V^*) - \delta U^*) + \delta V^*,$$

or shares 0 and risks the elite's challenge, which render the expected discounted payoff of

$$(1 - \pi(\phi^* + (1 - \phi^*)p))(1 + \delta V^*).$$

The ruler chooses the option that yields a greater discounted payoff, which implies that

$$(4) \quad V^* = \max \left\{ \begin{array}{l} 1 - ((1 - \phi^*)p(1 + \delta V^*) - \delta U^*) + \delta V^*, \\ (1 - \pi(\phi^* + (1 - \phi^*)p))(1 + \delta V^*) \end{array} \right\}.$$

Moreover, in equilibrium, the ruler chooses to share enough rents with the elite to make the latter indifferent if and only if

$$(5) \quad 1 - ((1 - \phi^*)p(1 + \delta V^*) - \delta U^*) + \delta V^* \geq (1 - \pi(\phi^* + (1 - \phi^*)p))(1 + \delta V^*).$$

First, suppose (5) holds, then the elite would give up challenging the ruler every time such an opportunity arises. As a result, the equilibrium is *peaceful* and the upper class is *stable* in the sense that the ruler and the elite can keep their status in the long run. In this case, the ruler shares the amount of

$$s^* = (1 - \phi^*)p(1 + \delta V^*) - \delta U^*$$

rents with the elite every period. As a result, the elite's discounted payoff and that of the ruler are,

respectively,

$$\begin{aligned} U^* &= (1 - \phi^*)p(1 + \delta V^*) \\ V^* &= 1 - ((1 - \phi^*)p(1 + \delta V^*) - \delta U^*) + \delta V^*. \end{aligned}$$

Solving the above system of equations about U^* and V^* yields

$$(6) \quad \begin{aligned} U^* &= \frac{(1 - \phi^*)p}{1 - \delta(1 - (1 - \delta)(1 - \phi^*)p)} \\ V^* &= \frac{1 - (1 - \delta)(1 - \phi^*)p}{1 - \delta(1 - (1 - \delta)(1 - \phi^*)p)}. \end{aligned}$$

Now suppose (5) fails, then the elite would challenge the ruler whenever there is such an opportunity. As a result, the equilibrium is *conflictual* and the upper class is *unstable* in the sense that the positions of the ruler and the elite are constantly changing among different individuals. In this case, the ruler shares no rent with the elite, so that

$$s^* = 0.$$

Because the elite would choose to challenge the ruler as long as $\theta = 1$, the elite's discounted payoff and that of the ruler are, respectively,

$$\begin{aligned} U^* &= (1 - \pi)\delta U^* + \pi(1 - \phi^*)p(1 + \delta V^*) \\ V^* &= (1 - \pi)(1 + \delta V^*) + \pi(1 - \phi^*)(1 - p)(1 + \delta V^*). \end{aligned}$$

Solving the above system of equations about U^* and V^* yields

$$(7) \quad \begin{aligned} U^* &= \frac{\pi}{1 - \delta(1 - \pi)} \frac{(1 - \phi^*)p}{1 - \delta(1 - \pi(\phi^* + (1 - \phi^*)p))} \\ V^* &= \frac{1 - \pi(\phi^* + (1 - \phi^*)p)}{1 - \delta(1 - \pi(\phi^* + (1 - \phi^*)p))}. \end{aligned}$$

2.4. Threat of Peasant Revolution and Power Centralization under Peace

It is meaningful to discuss power sharing, or inversely, power centralization, between the elite and the ruler only in peaceful equilibrium for which a stable upper class exists in the long run. There are two types of peaceful equilibrium depending on whether the upper class faces the threat of peasant

revolution, $\phi^* = 0$ or $\phi^* = \phi$.

Consider a peaceful equilibrium with no threat of peasant revolution, that is, one in which $\phi^* = 0$. Given that the equilibrium is peaceful, the discounted payoffs of the elite and that of the ruler, U^* and V^* , must satisfy (6). Taking $\phi^* = 0$ into (6) yields

$$(8) \quad \begin{aligned} U^* &= \frac{p}{1 - \delta(1 - (1 - \delta)p)} \\ V^* &= \frac{1 - (1 - \delta)p}{1 - \delta(1 - (1 - \delta)p)}. \end{aligned}$$

Two conditions must be met for the above U^* and V^* to be values of the elite and that of the ruler in a peaceful equilibrium with no peasant revolution. First, assuming $\phi^* = 0$, they must satisfy inequality (5), so that the equilibrium is peaceful. This requires

$$1 - (p(1 + \delta V^*) - \delta U^*) + \delta V^* \geq (1 - \pi p)(1 + \delta V^*).$$

Taking the U^* and V^* in (8) into the above inequality yields the first condition

$$\pi \geq 1 - \delta.$$

Second, the condition for $\phi^* = 0$ in (1) must hold for the value of V^* in (8), so that in the equilibrium the upper class need not face the threat of peasant revolution. This leads to the second condition

$$k \geq 1 + \delta \frac{1 - (1 - \delta)p}{1 - \delta(1 - (1 - \delta)p)} = \frac{1}{1 - \delta} \frac{1}{1 + \delta p}.$$

Now consider a peaceful equilibrium with the threat of peasant revolution, that is, one in which $\phi^* = \phi$. Given that the equilibrium is peaceful, the discounted payoffs of the elite and that of the ruler, U^* and V^* , must satisfy (6). Taking $\phi^* = \phi$ into (6) yields

$$(9) \quad \begin{aligned} U^* &= \frac{(1 - \phi)p}{1 - \delta(1 - (1 - \delta)(1 - \phi)p)} \\ V^* &= \frac{1 - (1 - \delta)(1 - \phi)p}{1 - \delta(1 - (1 - \delta)(1 - \phi)p)}. \end{aligned}$$

Similarly as before, two conditions must be met for the above U^* and V^* to be values of the elite and that of the ruler in a peaceful equilibrium with the threat of peasant revolution. First, assuming $\phi^* = \phi$, they must satisfy inequality (5), so that the equilibrium is peaceful. This requires

$$1 - ((1 - \phi)p(1 + \delta V^*) - \delta U^*) + \delta V^* \geq (1 - \pi(\phi + (1 - \phi)p))(1 + \delta V^*).$$

Taking the U^* and V^* in (9) into the above inequality yields the first condition

$$\phi \geq \frac{\left(\frac{1-\delta}{\pi} - 1\right)p}{1 + \left(\frac{1-\delta}{\pi} - 1\right)p}.$$

Second, the condition for $\phi^* = \phi$ in (1) must hold for the value of V^* in (9), so that in the equilibrium the upper class indeed has to face the threat of peasant revolution. This leads to the second condition

$$k \leq 1 + \delta \frac{1 - (1 - \delta)(1 - \phi)p}{1 - \delta(1 - (1 - \delta)(1 - \phi)p)} = \frac{1}{1 - \delta} \frac{1}{1 + \delta(1 - \phi)p}.$$

Proposition 1. 1. *If $\pi \geq 1 - \delta$ and $k \geq \frac{1}{1-\delta} \frac{1}{1+\delta p}$, then there exists a peaceful equilibrium with no threat of peasant revolution and in this equilibrium,*

$$s^* = \bar{s} := \frac{p}{1 + \delta p}.$$

2. *If $\phi \geq \frac{\left(\frac{1-\delta}{\pi} - 1\right)p}{1 + \left(\frac{1-\delta}{\pi} - 1\right)p}$ and $k \leq \frac{1}{1-\delta} \frac{1}{1 + \delta(1 - \phi)p}$, then there exists a peaceful equilibrium with the threat of peasant revolution and in this equilibrium,*

$$s^* = \underline{s}(\phi) := \frac{(1 - \phi)p}{1 + \delta(1 - \phi)p}.$$

3. *If $\pi \geq (1 - \delta)$ and $\frac{1}{1-\delta} \frac{1}{1+\delta p} \leq k \leq \frac{1}{1-\delta} \frac{1}{1 + \delta(1 - \phi)p}$, then two peaceful equilibria, one with the threat of peasant revolution and one with no such threat, coexist.*

Proposition 1 summarizes the above findings and characterizes conditions under which the two types of peaceful equilibrium exist. First, the existence of a peaceful equilibrium with no threat of peasant revolution requires the elite to be able to challenge the ruler sufficiently frequently, $\pi \geq 1 - \delta$, and the cost of launching a peasant revolution to be sufficiently high, $k \geq \frac{1}{1-\delta(1-(1-\delta)p)}$. The former condition

guarantees that the ruler is willing to share enough rents with the elite, preemptively appease potential challenges. This ensures the peace between the ruler and the elite. The latter condition guarantees that once arises, the peasant hero would not launch a peasant revolution against the upper class, being deterred by the prohibitively high cost of doing so. In this equilibrium, the amount of rents the ruler shares with the elite is $\bar{s}$ as defined in Proposition 1. Because the equilibrium is peaceful, the ruler and the elite just divide up the pool of rents at each period. In particular, the ruler's discounted payoff is exactly

$$\frac{1 - \bar{s}}{1 - \delta}.$$

The larger $\bar{s}$ is, the more power the ruler shares with the elite, the worse off the ruler is, and the lower the degree of power centralization is in this equilibrium.

Second, the existence of a peaceful equilibrium with the threat of peasant revolution requires the peasant hero to arise with a sufficiently large probability following the division within the upper class, $\phi \geq \frac{(\frac{1-\delta}{\pi}-1)p}{1+(\frac{1-\delta}{\pi}-1)p}$, and the cost of launching a peasant revolution to be sufficiently low, $k \leq \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p}$. Because

$$\frac{d}{d\phi} \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p} = \frac{1}{1-\delta} \frac{\delta p}{(1+\delta(1-\phi)p)^2} > 0,$$

both conditions hold when ϕ is sufficiently large. To understand the intuition behind these conditions, note that the amount of rents the ruler shares with the elite each period in this equilibrium is $\underline{s}(\phi)$ as defined in Proposition 1 and

$$\underline{s}'(\phi) = \frac{-p}{(1+\delta(1-\phi)p)^2} < 0.$$

This implies that a stronger threat of peasant revolution, in the sense that a peasant hero is more likely to arise following the division within the upper class, enables the ruler to share less rents with the elite to maintain peace. Knowing that challenging the ruler would risk a larger probability of having a peasant revolution that eliminates the entire upper class, the elite would be more reluctant to challenge the ruler and consequently willing to accept a smaller amount of rents to give up challenging the ruler. Because a larger ϕ enables the ruler pay less for rents, two effects follow. On the one hand, peace is more affordable for the ruler, so that when ϕ is sufficiently large, the condition for peaceful equilibrium (5) can be sustained. This is the intuition behind the condition $\phi \geq \frac{(\frac{1-\delta}{\pi}-1)p}{1+(\frac{1-\delta}{\pi}-1)p}$. On the other hand, in the peaceful equilibrium with the threat of peasant revolution, the ruler is better off when the

threat grows stronger. Formally, because the ruler shares $\underline{s}(\phi)$ with the elite at each period, the ruler's discounted payoff is

$$\frac{1 - \underline{s}(\phi)}{1 - \delta},$$

which clearly is strictly increasing in ϕ . This implies that the larger ϕ is, the more attractive becoming a ruler would be for the peasant hero facing a fixed cost of launching a peasant revolution. This is the intuition for why the second condition for the existence of a peaceful equilibrium with the threat of peasant revolution, $k \leq \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p}$, also requires a sufficiently large ϕ .

It is worth comparing the peaceful equilibrium with no threat of peasant revolution and the peaceful equilibrium with such threat. Note that internal peace within the upper class endures in both equilibria. Hence, in both equilibria, the ruler and the elite divide up the whole rents at each period, generating no deadweight loss. As a result, the two types of peaceful equilibrium cannot be Pareto ranked. The only difference is how the rents are divided between the ruler and the elite. Because

$$\underline{s}(\phi) < \bar{s}$$

holds for all $\phi \in (0, 1)$, the ruler always shares less rents with the elite and therefore power is always more centralized to the ruler in the peaceful equilibrium with the threat of peasant revolution in comparison to the peaceful equilibrium with no such threat. The threat of peasant revolution facilitates power centralization under peace. Moreover, because $\underline{s}(\phi)$ is strictly decreasing in ϕ , the stronger the threat of peasant revolution is, the more the ruler can exploit such threat to centralize power.

According to Proposition 1, the two types of peaceful equilibrium coexist when the elite can get opportunities to challenge the ruler sufficiently often, $\pi \geq 1 - \delta$, and when the cost of peasant revolution is intermediate,

$$\frac{1}{1-\delta} \frac{1}{1+\delta p} \leq k \leq \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p}.$$

Under these conditions, the presence and absence of the threat of peasant revolution are *self-fulfilling prophecies*. On the one hand, suppose the upper class believes that if it is divided, there is a possibility that a peasant hero would rise to launch a revolution that destroys the entire upper class. Fearing the risk of peasant revolution, the elite demands less power sharing to give up challenging the ruler.

Exploiting the elite's fear, the ruler is able to centralize power without disturbing peace, sharing only $\underline{s}(\phi)$ with the elite, preserving a considerable amount $1 - \underline{s}(\phi)$. Because the ruler's power is sufficiently centralized, becoming a ruler is sufficiently profitable. Then, because revolution is not too costly,

$$k \leq \frac{1}{1 - \delta} \frac{1}{1 + \delta(1 - \phi)p},$$

once arising, the peasant hero is willing to launch a revolution with the purpose of eliminating the upper class and becoming the ruler. The belief of the upper class about the threat of peasant revolution therefore can be confirmed. On the other hand, suppose the upper class believes that even if it is divided, the worst possible case is an internal conflict — there would be no chance of any peasant revolution. Then, the elite would require a high level of power sharing for not challenging the ruler when such an opportunity arises. As a result, for peace to be maintained, the ruler's power cannot be too much centralized. The ruler must share a considerable amount $\bar{s}$ with the elite, preserving only $1 - \bar{s}$. Because the ruler's power cannot be centralized, becoming a ruler is not that profitable. Then, because revolution is sufficiently costly,

$$k \geq \frac{1}{1 - \delta} \frac{1}{1 + \delta p},$$

once arising, the peasant hero would prefer not to launch a revolution. The belief of the upper class about the nonexistence of the threat of peasant revolution therefore can also be confirmed, just like the opposite belief.

Therefore, coordination matters when two types of peaceful equilibrium coexist. Culture, history, or shared memories in this case are crucial for which equilibrium would become the focal point that the upper class coordinates on. Historical records or even folk tales of destructive peasant revolutions may contribute to the emergence of a peaceful equilibrium with the threat of peasant revolution in which the ruler's power is more centralized.

2.5. *Threat of Peasant Revolution and Peace*

According to Proposition 1, peaceful equilibrium with no threat of peasant revolution does not exist when $\pi < 1 - \delta$. Hence, when the elite only very infrequently has opportunities to challenge the ruler,

internal peace within the upper class is impossible without the threat of peasant revolution. They can only be sustained in a peaceful equilibrium with the threat of peasant revolution, which necessitates a sufficiently large probability of the rise of a peasant hero following the division of the upper class,

$$\phi \geq \frac{\left(\frac{1-\delta}{\pi} - 1\right)p}{1 + \left(\frac{1-\delta}{\pi} - 1\right)p} \in (0, 1),$$

and a sufficiently low cost of peasant revolution, $k \leq \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p}$. The interesting question is whether the lack of the threat of peasant revolution combined with persistent internal conflict within the upper class can be sustained as an equilibrium.

Assume $\pi < 1 - \delta$ and consider a conflictual equilibrium with no threat of peasant revolution, in which $\phi^* = 0$. Given that the equilibrium is conflictual, the discounted payoffs of the elite and that of the ruler, U^* and V^* , must satisfy (7). Taking $\phi^* = 0$ into (7) yields

$$(10) \quad \begin{aligned} U^* &= \frac{\pi}{1 - \delta(1 - \pi)} \frac{p}{1 - \delta(1 - \pi p)} \\ V^* &= \frac{1 - \pi p}{1 - \delta(1 - \pi p)}. \end{aligned}$$

Two conditions must be met for the above U^* and V^* to be values of the elite and that of the ruler in a conflictual equilibrium with no peasant revolution. First, assuming $\phi^* = 0$, they must fail (5), so that the equilibrium is conflictual. This requires

$$1 - (p(1 + \delta V^*) - \delta U^*) + \delta V^* < (1 - \pi p)(1 + \delta V^*),$$

so that the ruler has no incentive to deviate to sharing enough rents with the elite to buy peace. Taking the U^* and V^* in (10) into the above inequality yields the first condition

$$\pi < \frac{1 - \delta}{1 - \delta(1 - \pi)},$$

which is implied by the assumption that $\pi < 1 - \delta$. Second, the condition for $\phi^* = 0$ in (1) must hold for V^* in (10), so that the threat of peasant revolution is absent in the equilibrium. This leads to the

second condition

$$k \geq 1 + \delta \frac{1 - \pi p}{1 - \delta(1 - \pi p)} = \frac{1}{1 - \delta(1 - \pi p)}.$$

Therefore, assuming $\pi < 1 - \delta$, the lack of the threat of peasant revolution and persistent internal conflict within the upper class can occur simultaneously as an equilibrium as long as the cost of peasant revolution is sufficiently high, so that $k \geq \frac{1}{1 - \delta(1 - \pi p)}$. According to Proposition 1, such a threshold level of cost is not too high to preclude even the peaceful equilibrium with the threat of peasant revolution if and only if

$$\frac{1}{1 - \delta(1 - \pi p)} \leq \frac{1}{1 - \delta} \frac{1}{1 + \delta(1 - \phi)p},$$

which can be simplified as

$$\phi \geq 1 - \frac{\pi}{1 - \delta} \in (0, 1).$$

Proposition 2. *If $\pi < 1 - \delta$, $\phi \geq 1 - \frac{\pi}{1 - \delta}$, and $\frac{1}{1 - \delta(1 - \pi p)} \leq k \leq \frac{1}{1 - \delta} \frac{1}{1 + \delta(1 - \phi)p}$, then two equilibria coexist: a peaceful equilibrium with the threat of peasant revolution and a conflictual equilibrium with no threat of peasant revolution.*

According to Proposition 2, a peaceful equilibrium exists with a conflictual equilibrium depending on whether the upper class faces the threat of peasant revolution under three conditions: (1) the elite very infrequently gets opportunities to challenge the ruler, $\pi < 1 - \delta$, (2) the probability that a peasant hero arises after the division of the upper class is sufficiently high, $\phi \geq 1 - \frac{\pi}{1 - \delta}$, and (3) the cost of peasant revolution is intermediate. When all these conditions hold, whereas the expectation of peasant revolution and a stable upper class in which the ruler's power is centralized reinforce each other into a peaceful equilibrium; the expectation of no peasant revolution and the persistent division within the upper class reinforce each other into another conflictual equilibrium. The threat of peasant revolution is not only essential to the ruler's power centralization, but also crucial for the existence of a long-run stable upper class. As shown in the discussion below Proposition 1, the threat of peasant revolution is self-fulfilling and supports a peaceful equilibrium in which the ruler centralizes power, because of the large probability of peasant revolution after internal conflict,

$$\phi \geq 1 - \frac{\pi}{1 - \delta} > \frac{\left(\frac{1 - \delta}{\pi} - 1\right)p}{1 + \left(\frac{1 - \delta}{\pi} - 1\right)p},$$

and the not too high cost of peasant revolution,

$$k \leq \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p}.$$

Now suppose the upper class instead expect no threat of peasant revolution after being divided. Then, the elite would require a high level of power sharing for maintaining peace when the opportunity of challenging the ruler arises. This means that peace is expensive for the ruler. Moreover, because the elite only very infrequently obtains any opportunity to challenge the ruler and because the ruler does not have to worry about peasant revolution even when being challenged, the ruler *ex ante* prefers to withhold any power sharing and fights the elite in an internal conflict when the latter is lucky to get an opportunity to challenge. At last, because the ruler constantly has to cope with internal conflicts with the elite, becoming a ruler is not that profitable. Then, because revolution is sufficiently costly,

$$k \geq \frac{1}{1-\delta(1-\pi p)},$$

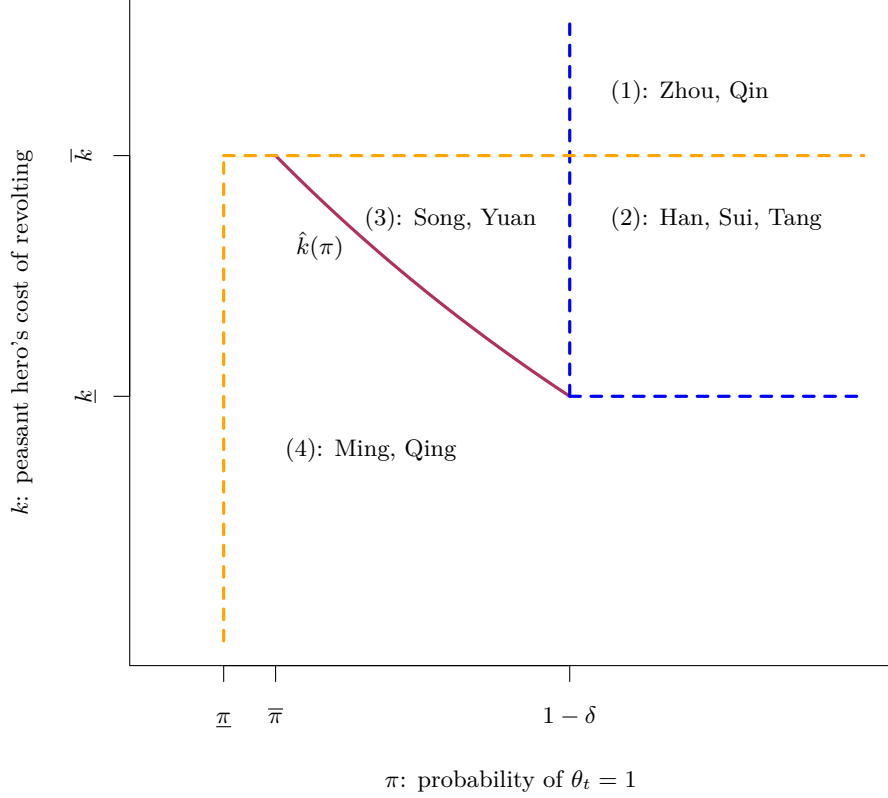
once arising, the peasant hero would given up launching any revolution. The expectation of the upper class about no threat of peasant revolution therefore is self-fulfilling.

3. ILLUSTRATION OF EQUILIBRIUM TRANSITION

Consider a hypothetical state. The analysis of multiple equilibria in Section 2 implies a potential trajectory for historical development. Although my model is not fully dynamic, let's assume that exogeneous parameters ϕ (the chance of peasant revolution) and k (the cost of peasant revolution), along with endogenous variable s^* (equilibrium power sharing), would continually evolve through rounds of peasant revolutions, subsequently resulting in a change in the exogeneous parameter π (the chance of the elite's challenge).

Area 1

Initially, the elite poses a significant threat to the ruler ($\pi \geq 1 - \delta$), resulting in a power-sharing arrangement; as there is no successful precedent, the cost of a peasant revolution is deemed prohibitively high ($k \geq \bar{k}$). The status quo is a peaceful equilibrium without threat of peasant revolution, represented by area (1) in Figure 1.



Notes: $\underline{\pi} := \frac{p}{p+\phi(1-p)}(1-\delta)(1-\phi)$, $\bar{\pi} := (1-\delta)(1-\phi)$, $\underline{k} := \frac{1}{1-\delta} \frac{1}{1+\delta p}$, $\bar{k} := \frac{1}{1-\delta} \frac{1}{1+\delta(1-\phi)p}$, $\hat{k}(\pi) := \frac{1}{1-\delta(1-\pi p)}$. Peaceful equilibrium without threat of peasant revolution exists to the upper right of the dashed blue lines; peaceful equilibrium with threat of peasant revolution exists to the lower right of the dashed yellow lines; the threat of peasant revolution is essential for peace in the area encircled by the solid red, dashed yellow, and dashed blue lines; the upper left corner consists only of conflictual equilibria, where no peaceful equilibria, or unified dynasties, exist.

Figure 1: Equilibria

Area 2

An exogenous shock occurs and raises the awareness that a peasant revolution can take place. Consequently, the actors quickly coordinate on a peaceful equilibrium with threat of peasant revolution, in which the ruler shares less power with the elite. The status quo transitions to coexistence of both peaceful equilibriums, represented by area (2), as the perceived cost of peasant revolution drops to a reasonable level ($\bar{k} \geq k \geq \underline{k}$) and the elite retains a significant chance to challenge the ruler ($\pi \geq 1 - \delta$).

Area 3

As the ruler consolidates more power from the elite due to the pressures of peasant revolution, this may ultimately lead to a drop in the elite's chance of challenging the ruler ($\pi < 1 - \delta$). The status quo transitions to a coexistence of peaceful equilibrium with threat of peasant revolution and conflictual equilibrium without threat of peasant revolution, represented by area (3). Although peace can still be maintained, the ruler shares even less power with the elite. In the event of a coordination failure, the ruler may end up sharing no power and engaging in conflict with the elite if the latter gets the chance to challenge, as this chance is so small that the ruler need not bother to appease the elite unless peasant revolution threatens the ruler's own interests.

One caveat of area (3) is that the revolutionary threat of a third party provides an alternative means by which a ruler can credibly commit to its contender. Acemoglu and Robinson (2000, 2001) argue that the modern Western world has developed democratic institutions to solve the commitment problem between the rich and the poor. As will be shown later, Chinese emperors successfully committed to sharing power with the elite and maintaining long-term stability due to the persistent pressures from peasant revolutions, even though the power shared in this case is limited.

Area 4

If the threat of peasant revolutions intensifies, further decreasing their perceived costs (k), or if power centralization continues, further decreasing the elite's chance to challenge (π), the status quo may transition to a peaceful equilibrium with threat of peasant revolution alone, represented by area (4). In this scenario, the ruler concedes only an infinitesimal amount of power to secure the compliance of the elite, and no conflict arises on the equilibrium path, as either the threat from the peasants is too

large or the threat from the elite is too small.

4. HISTORICAL PERSPECTIVES

In this section, I will present historical evidence from Imperial China. Spanning thousands of years, Chinese history witnesses the rise and fall of numerous dynasties, which can be challenging to follow. I will attempt to present this history with interpretations from the model to make it more accessible.

Chinese history unfolds in a manner closely resembling the transition in Section 3. Initially, the Zhou dynasty maintained a quasi-feudal system, where nobles, with hereditary fiefs and independent military capacity, posed a significant threat to the ruler, resulting in a power-sharing structure. However, a series of events, including the Zhou rulers' introduction of the Mandate of Heaven and the pivotal role of peasant revolution in the fall of Qin and the establishment of Han, rapidly propagated a new political culture. This culture suggested that peasants held both the legitimacy and capability to claim rulership and replace the entire upper class. Subsequent dynasties, including the Han, Sui, and Tang, witnessed an increasing level of power centralization due to the growing threat of peasant revolution, though aristocrats still retained a decent amount of power. The ongoing erosion of elite influence reached a critical point during the Tang-Song transition, as the imperial examination system solidified its position as the primary avenue for personnel selection. This marked a transformative shift, with the elite class now composed of scholarly bureaucrats rather than aristocrats. Consequently, their chance of posing challenges to the ruler significantly diminished, and their roles were greatly diminished to that of mere administrative staff for the ruler. As peasant rebellions continued to shape the course of Chinese history, both in establishing and toppling the Ming dynasty, the rulers of the last two dynasties, the Ming and Qing, effectively approached the pinnacle of power centralization in Chinese Imperial history.

Below, I will conduct a comprehensive review of historical events to substantiate the narrative presented earlier. In Section 4.1, I delve into the origins of the political philosophy of the Mandate of Heaven around 1000 BC, tracing its evolution over time and emphasizing its impact on dynastic changes and political legitimacy in the following millennia. In Section 4.2, I provide a chronological overview of relevant major developments in Chinese political history. I present a table outlining key

events marking the consolidation of political power, along with significant instances of peasant rebellions or peasant-led military regimes. This juxtaposition highlights the intricate interplay between the evolution of power centralization and the unfolding of peasant rebellions. I also supplement this with a more detailed discussion of crucial events and turning points. Finally, I conclude with my observations on the underlying trend that has shaped the extensive course of Chinese history.

4.1. Political Legitimacy in Historical Perspectives

In stark contrast to Medieval Europe, ancient China lacked a systematic religious or clerical institution to confer legitimacy upon ruling authorities. Therefore, when King Wu established the Zhou dynasty through military conquest over the Shang dynasty in 1046 BC, the Zhou rulers faced the pressing issue to legitimize their rule. They ingeniously devised a political philosophy known as the “Mandate of Heaven,” in which heaven would bestow upon a ruler the divine mandate based on their moral virtue and righteous governance over the people. However, this entitlement was not unshakeable; if a ruler failed to fulfill his duties and rendered his people to suffer, exemplified by the last king of the Shang dynasty, the mandate would be retracted, leaving the ruler vulnerable to rival claimants who aimed to seize the throne. By embracing this philosophy, the Zhou rulers had to recognize the unsettling reality that any ruling faction, including their own, could face ousting from power if they behaved improperly. The “Book of Odes,” a literary masterpiece composed during the Zhou period, clearly underscored this cautionary principle.

This philosophy experienced further development during the Han dynasty (206 BC-220), when rulers sought to centralize power and needed justification for their aggrandizement. Emperor Wu decided to endorse Dong’s ideology of “interactions between heaven and mankind”, which integrated the Mandate of Heaven with yin-yang cosmologies. This ideology not only justified the imperial system within the natural order of the universe but also established criteria to evaluate regime legitimacy and imposed constraints on rulers. More specifically, heaven would bestow a favorable natural order as a sign of approval when emperors ruled with virtue and righteousness, and would express its displeasure through natural disasters like floods and famines when observing unsatisfactory governance. In the latter situation, the emperor was expected to acknowledge and rectify his mistakes and shortcomings to preserve his legitimacy and prevent potential revolts. Throughout subsequent history, Chinese

emperors widely accepted this ideology and issued self-critical edicts in response to calamities caused by human or natural factors, seeking forgiveness from Heaven. Sometimes, however, the theory was invoked retrospectively, i.e., after a dynasty had been toppled, as evidence that the ruler was unworthy and had lost the mandate.

This political philosophy has left distinct imprints on Chinese history, which sets up stage for my model. First, it facilitated the cyclical overturn of dynasties. Historically, as it implied the right of rebellion against unjust rulers, it helped legitimize peasant rebellions to overthrow old regimes and substantiate the claims of new dynasties. Certainly, the right of rebellion was not coded into any official law; rather, it remained a politically sensitive topic in every dynasty. Nevertheless, it was still considered a positive right grounded in the Chinese moral system. Second, this philosophy encouraged peasants to actively participate in politics from an early stage and in an extensive manner. Its meritocratic nature, which prioritized capable governance over noble lineage, greatly benefited commoner rebels such as Liu, a low-ranking officer who founded the Han dynasty (206 BC-220), and Zhu, a penniless peasant who founded the Ming dynasty (1368-1644). Despite their humble beginnings, these dynasties are celebrated as among the most successful in Chinese history, unequivocally attaining the divine mandate. My model incorporates these distinctive features of Chinese history in that it iterates itself and allows the peasant hero to assume rulership in the subsequent period following a revolution. Finally, this philosophy arguably incentivized rulers to deliver good governance. It has been frequently invoked as the sole effective check upon the abuse of power by rulers in a system that had few other constraints and has significantly influenced policy debates throughout history. For instance, some historians contend that the Ming and Qing dynasties' adoption of a conservative fiscal system, aimed at alleviating the taxation burden on the populace, was mainly incentivized by the traumatic experiences of peasant revolutions during previous dynastic changes³ (Zhang, 2023).

4.2. Power Centralization and Peasant Rebellion

The history of Imperial China follows a cyclical pattern of dynastic rise and fall, punctuated by intervals of warring division. An interesting pattern emerges, where after a tumultuous period of

³However, these policies gave rise to other issues beyond this initial concern and potentially led to China's relative economic decline in the 18th and 19th centuries.

Table 2: Power Centralization in Imperial China

Dynasty	Peasant-led Violence	Power Centralization	Elite Selection
Ancient China			
Zhou (1046-256 BC)		- Feudal system	Hereditary system
- Warring States (475-221 BC)		- Introduce Mandate of heaven	
Early Imperial China			
Qin (221-206 BC)		- Promotion based on merit	Hereditary system to
		- County-commandery system	Recommendation system
	Chen Sheng Wu Guang Uprising (209-208 BC)		
	- facilitate Qin's fall		
	Liu Bang Group Rebellion (209-202 BC)		
	- establish Han		
Han (206 BC-220)		- Phase out feudal legacy	Recommendation system
		- Endorse Mandate of Heaven	
		- Endorse Confucianism	
		- State monopoly on salt/iron	
	Yellow Turban Rebellion (184-205)		
	- facilitate Han's fall		
	Northern and Southern Dynasties (420-589)		
	- military regimes		
Mid Imperial China			
Sui (581-618)		- Introduce Imperial Exam	Recommendation system
		- Equalize land distribution	to Exam system
		- Restructure bureaucracy	
		(five departments six boards)	
Tang (618-907)			Recommendation system
	Huang Chao Rebellion (875 - 884)		to Exam system
	- overthrow Tang		
	Five Dynasties and Ten Kingdoms (907-960)		
	- military regimes		
Late Imperial China			
Song (960-1279)		- Consolidate Imperial Exam	Exam system
		- Centralize control on military	
		(bureaucrat oversee military)	
Yuan (1271-1368)			
	Red Turban Rebellion (1351-1368)		
	- establish Ming		
Ming (1368-1644)		- Abolish chief bureaucrat	Exam system
		- Establish secret police	
		- Constrain local power	
		(san si system, etc.)	
		- Eunuch as balance	
	Li Zicheng Rebellion (1629-1644)		
	- overthrow Ming		
Qing (1636-1912)		- Machu aristocrat as balance	Exam system
		- Confidential memorial system	
	Taiping Rebellion (1850-1864)		
	- facilitate Qing's fall		

peasant-led violence, the succeeding dynasty often undergoes a new phase of power centralization. Table 1 provides a chronological overview of notable events related to both power centralization and peasant-led violence. Further details are provided below.

Ancient China. Around 1046 BC, Wu, a former vassal of the Shang dynasty, defeated King Zhou of Shang in a battle and established the Zhou dynasty. Upon its establishment, the conquered territory was divided into hereditary fiefs, mirroring the decentralized feudal systems observed in Medieval Europe. In the absence of an established religion and clerical institution, the Zhou rulers introduced the philosophy of the Mandate of Heaven to legitimize their overtaking the Shang. Over time, peripheral territories gained power on par with that of the Zhou, resulting in a decline of central authority. A succession of states claimed independence and engaged in a competition for dominance.

Early Imperial China. The Qin state, a lower-ranking fief of Zhou, completed the conquest of other warring states and unified the country. Qin stood out because they implemented a series of centralization reforms, including the promotion of ministers based on merit rather than bloodline. These efforts culminated in the introduction of the county-commandery system, aiming to centralize local appointments and dismantle the feudal system. However, these reforms were ahead of their time and faced strong resistance, leading to the quick collapse of the Qin dynasty. Given my model, this collapse was likely because the status quo was still in a peaceful equilibrium without threat of peasant revolution, and the ruler was expected to share a significant amount of power. Things began to change as two peasants, Chen and Wu, spearheaded widespread uprisings against the Qin. They famously challenged the legitimacy of hereditary privileges, questioning, “Are kings and nobles given their high status by birth?” Liu, a descendant of a peasant family, emerged triumphant over all rivals and established the Han dynasty. The status quo rapidly shifted to a peaceful equilibrium with threat of peasant revolution, enabling the rulers to seize more power from the elites. Starting with feudal remnants like semi-autonomous kingdoms, the Han dynasty progressively transitioned them into centralized commanderies. The Han rulers also strengthened the ideological foundation by embracing Dong’s cosmology and Confucianism, established a personnel selection system based on recommenda-

tion⁴, and nationalized the salt and iron industries to fund the military campaigns. The Han dynasty endured for almost 400 years in relative peace but eventually declined amid internal conflicts, creating an opening for peasant-led uprisings such as the Yellow Turban Rebellion. These triggered an almost four-century-long era of warfare, where a multitude of military regimes, led by warlords from humble backgrounds, ascended, placing the upper social stratum in a precarious position.

Tang-Song Transition Yang, a member of a military faction in northwestern China, brought an end to this tumultuous period and established the Sui dynasty. The escalating threat of peasant-led violence enabled the Sui rulers to implement a series of reforms aimed at centralizing power. These included the equal-field system to redistribute land to peasants, the Five Departments and Six Boards system to bureaucratize the central administration, and efforts to introduce the imperial examination system. The reforms jeopardized the privileges of the aristocracy, who still retained a significant share of power, and led to the removal of the second Sui ruler by a member within the same military faction. This internal transition hindered further progress in power centralization, as the Tang dynasty inherited most of the institutions from the Sui dynasty. The Tang dynasty was another lengthy dynasty that lasted for almost 300 years. However, at its height, the Tang dynasty faced internal challenges initiated by regional commanders, leading to growing turmoil, among which a peasant rebellion led by Huang Chao delivered a fatal blow to the dynasty. Subsequently, Chinese history plunged into another dark period of political upheaval marked by a rapid succession of military regimes known as the Five Dynasties and Ten Kingdoms. The once influential aristocratic families of the Sui and Tang dynasties were largely marginalized during this turbulent time.

Some historians argue that Chinese history underwent a major shift before and after this turbulent time, known as the Tang-Song transition⁵. On the political front, aristocracies were replaced by scholarly bureaucrats, and political power became increasingly concentrated in the hands of the ruler. While scholarly bureaucrats as a group still retained certain advantages in society, their titles were no longer hereditary. According to my model, this occurred because the decline in the elite's power

⁴Initially, the recommendation system demonstrated greater meritocracy and progressiveness compared to the hereditary approach. However, it gradually devolved into a closed game dominated by a limited circle of landed aristocrats and military aristocrats. Concerned that the selection of elites and the related political power would fall entirely under the control of this privileged group, the subsequent dynasties introduced the imperial exam as a replacement.

⁵Modern scholarship in this area begins with Japanese scholars like Naitō Konan and Katō Shigeshi.

finally led to a drop in the exogenous factor π , thereby shifting the status quo to a coexistence of peaceful and conflictual equilibrium where the elites wield far less power. In addition to consolidating the imperial examination system as the primary method for personnel selection, the Song rulers also centralized control over the military by appointing bureaucrats to oversee military operations⁶. This practice proved to be perilous when confronting threats from northern nomadic regimes, and the Song dynasty was ultimately defeated by the Mongols. The Yuan rulers attempted to replicate centralized institutions, but their efforts were thwarted by resistance from Mongolian aristocrats. According to my model, at this stage of the game, internal conflict would break out if there is no consensus about the threat of peasant revolutions among elites. This coordination failure soon exacted a price, as the internal power struggle led to peasant rebellions such as the Red Turban Rebellion, which rapidly spread and ultimately toppled the Yuan dynasty.

Ming and Qing. As peasant revolutions successfully established and overthrew the Ming dynasty, and the perceived cost of peasant revolution decreased below a threshold, the status quo finally transitioned to a peaceful equilibrium with threat of peasant revolution alone. In this scenario, the ruler shares an infinitesimal amount of power to ensure the compliance of the elites, and this structure is highly stable, free from internal conflicts. The last two dynasties, the Ming and Qing, closely fit into this narrative, each lasting about 300 years, and witnessed the zenith of imperial power. The founder of the Ming dynasty orchestrated the most comprehensive reforms to centralize power, resulting in an almost one-man rule. His reforms included abolishing the chief bureaucrat to assume the role himself, establishing a secret police to carry out purges, and imposing strict checks and balances at local level. The later Ming rulers established a personal advisory office to assist them with state affairs without compromising their control. Furthermore, they employed eunuchs to balance the influence of scholarly bureaucrats, enabling them to navigate these tensions and maintain their position as the ultimate arbitrators. However, the growing internal strifes exacerbated external problems such as social unrest and the Manchu invasion. The capital fell to a peasant rebel army led by Li Zicheng, who ultimately lost to the Manchurians. The Qing dynasty further consolidated power based on Ming's

⁶A noteworthy incident from this era lends credence to my theory, referred to as the “dismissal of military power using wine cups.” In 961, the founding Emperor of the Song Dynasty hosted a banquet, subtly suggesting to his senior army commanders that they voluntarily surrender their military authority in exchange for a peaceful and comfortable life. Unwilling to confront the uncertainty of conflict, these commanders accepted the proposal, resigning from their military positions the very next day.

legacies. The Qing rulers employed Manchu aristocrats to balance scholarly bureaucrats, which not only prevented the aristocrats from reverting to their tribal tradition of collective decision-making, unlike in the Yuan dynasty, but also eliminated the problems associated with using eunuchs, unlike in the Ming dynasty. They also devised the confidential memorial system, facilitating direct information access to the throne. Throughout their reign, the Qing rulers effectively retained power, but eventually succumbed to a combination of peasant rebellions, foreign invasions, and Republican revolutions.

Summary. Throughout Chinese history, there has been a dual trend of increasing recognition of peasants' claim to political power and a simultaneous trajectory of power centralization. The notion that any man could rise to nobility regardless of his bloodline took root during the peasant rebellion in 209 BC, led by Chen Sheng and Wu Guang, and soon found embodiment in Liu Bang, a peasant who founded the Han dynasty, which ultimately abolished the feudal system. Since then, centralized imperial states came to dominate Chinese history, and feudal lords completely faded from the scene. Subsequently, peasant rebellions played a crucial role in both founding and overthrowing dynasties. Examples include the Yellow Turban Rebellion overthrowing the Han dynasty, the Huang Chao Rebellion overthrowing the Tang dynasty, the Red Turban Rebellion overthrowing the Yuan dynasty and founding the Ming dynasty, and the Li Zicheng Rebellion overthrowing the Ming dynasty. Peasant-led military regimes also contributed to prolonged periods of division, as exemplified by the Northern and Southern dynasties and the Five Dynasties and Ten Kingdoms. After each significant period of peasant-led violence, the succeeding dynasty demonstrated a strong capacity to enhance power centralization. This process culminated in the Ming and Qing dynasties, which was facilitated by a highly systemized examination process and the high replaceability of bureaucrats. Conversely, dynasties established through internal takeover or nomadic conquest, such as the Eastern Han, Tang, and Yuan, showed less inclination for further centralization. The intertwined development of peasant revolution and power centralization emerges as a prominent feature of Chinese history.

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